LRP: CONFLUENCE-AWARE PROFILING OF PARAMAGNETIC RIM LESIONS

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BACKGROUND

CALs: an emerging therapeutic target[1]

- Current therapies effectively reduce relapse-associated inflammation
- Yet disability still progresses independently of relapses (PIRA)
- PIRA is driven by smoldering inflammation within chronic active lesions (CALs)

PRLs as an in-vivo surrogate[2]

- CALs show a rim of iron-laden microglia surrounding a demyelinated core
- Susceptibility MRI detects this rim in vivo, defining paramagnetic rim lesions (PRLs)
- Rim and core of the lesions are histologically distinct cellular profiles[3]

The gap

- PRLs can be confluent with neighbouring lesions
- No automated method leverages susceptibility contrast to isolate PRLs
- Compartment-level (core vs rim) analysis therefore remains impractical at scale

OBJECTIVES

1. Develop the **Lesion Radial Profiler (LRP)**, an automated method to separate confluent PRLs.
2. Explore its use for radial profiling (core, rim, perilesional) with quantitative MRI.

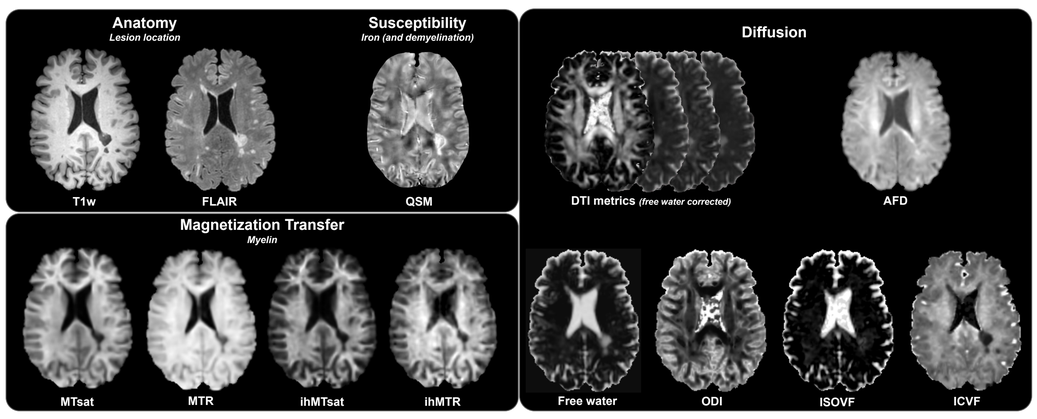


Figure 1. Multimodal MRI data[4,5]. Representative axial slices from one participant. Anatomy: T1-weighted (T1w) and fluid-attenuated inversion recovery (FLAIR). Susceptibility: quantitative susceptibility mapping (QSM). Magnetization transfer: magnetization transfer saturation (MTsat) and ratio (MTR); inhomogeneous magnetization transfer saturation (ihMTsat) and ratio (ihMTR). Diffusion: free-water-corrected diffusion tensor imaging (DTI) metrics (fractional anisotropy, FA; mean, axial and radial diffusivity, MD, AD, RD); neurite orientation dispersion and density imaging (NODDI) metrics (intracellular volume fraction, ICVF; orientation dispersion index, ODI; isotropic volume fraction, ISOVF); free water fraction (FW); and apparent fibre density (AFD).

CONCLUSION

➔ **LRP enables compartment-level analysis of confluent PRLs**
➔ Next steps: full automation of PRL identification, longitudinal PRL tracking

Limitations: semi-automatic (expert PRL identification), small cohort (n = 20), cross-sectional; no reference for the separation and rim segmentation steps

github.com/zinelya/LRP

METHODS

Participants and MRI
20 persons with relapsing-remitting MS underwent 3T MRI (Fig. 1)

Preprocessing
Images were preprocessed with standard pipelines and coregistered to T1w space

Lesion analysis
Confluent PRLs were separated and profiled with LRP (Fig. 2)

Statistics
Linear mixed-effects models*; FDR-corrected contrasts*

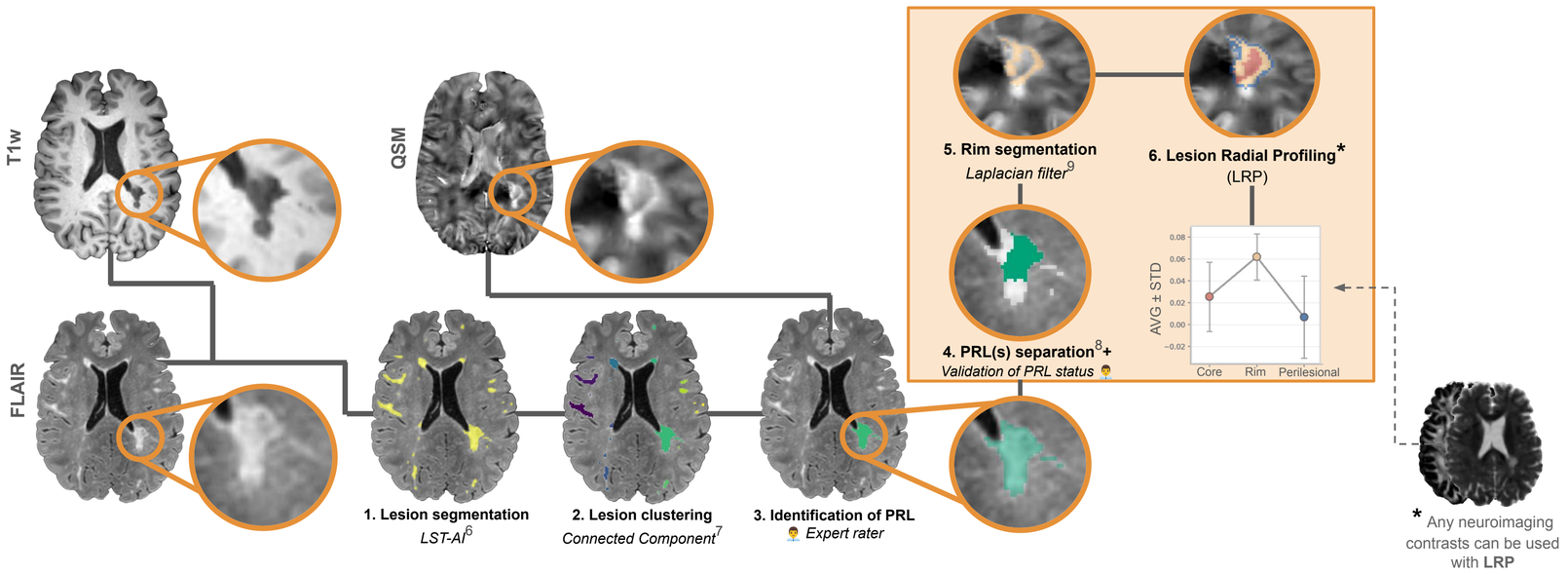


Figure 2. Lesion Radial Profiler (LRP) pipeline. Lesions were segmented on FLAIR and T1w using LST-AI and grouped into connected components. An expert rater identified PRL-containing clusters, which were separated into individual lesions with a watershed combining quantitative susceptibility mapping (QSM), FLAIR and T1w. PRL status was validated, the paramagnetic rim was segmented on QSM (Laplacian edge detection). The *core* was defined as the lesion interior enclosed by the rim; the *peri-lesional* compartment was a 2-mm shell around the lesion, derived from a distance map and restricted to white matter, excluding other lesions. Quantitative metrics were then extracted in each compartment.

RESULTS

20 persons with RRMS → **1,475** lesions → **66** PRLs → **42** confluent PRLs **(63.6%)**

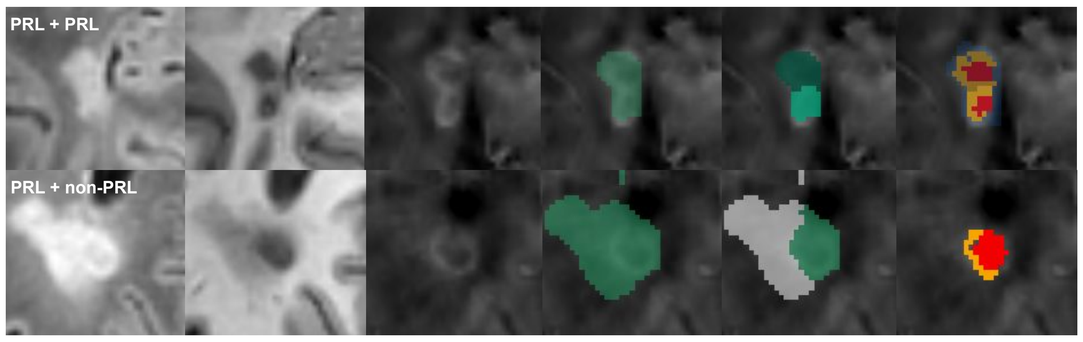


Figure 3. Separation of confluent PRLs. Top: two adjacent PRLs. Bottom: a PRL merged with a non-PRL lesion. Automated segmentation yields one label per cluster; LRP separates individual lesions.

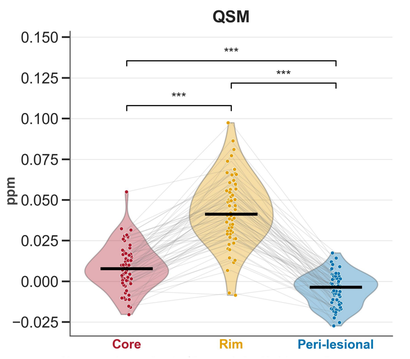


Figure 4. QSM across compartments. QSM, the contrast used to define the rim, is highest in the rim. Lines join compartments of the same lesion; bars = medians; *** FDR-corrected p < 0.001.

	QSM	MTsat	MTR	ihMTsat	ihMTR	FA	MD	AD	RD	AFD	Free water	ISOVF	ICVF	ODI
Rim − Peri	+1.90***	-1.27***	-1.05***	-1.11***	-0.66***	-0.44***	+1.05***	+0.80***	+1.07***	-0.18***	+1.12***	+0.22	-1.22***	-0.17
Core − Rim	-1.36***	-0.38***	-0.38***	-0.14	-0.05	-0.14	+0.53***	+0.50***	+0.51***	+0.07	+0.50***	+0.00	-0.63***	-0.36***
Core − Peri	+0.54***	-1.65***	-1.43***	-1.26***	-0.70***	-0.58***	+1.58***	+1.29***	+1.58***	-0.11*	+1.62***	+0.22	-1.85***	-0.53***

−2.0 −1.5 −1.0 −0.5 0.0 0.5 1.0 1.5 2.0

Figure 5. Compartment contrasts, all metrics. Standardized differences (SD) from linear mixed models; *FDR p < 0.05, ** p < 0.01, *** p < 0.001.

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